

TrES-2b

R-2430

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110701 · created 2026-08-22T00:30:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455743.888846 +/- 9.1e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.014
Transit depth (Rp/Rs)^2	1.16 +/- 0.3 [%]
Orbital Inclination (inc)	84.15 +/- 1.7
Airmass coefficient 1 (a1)	1.0207 +/- 0.0072
Airmass coefficient 2 (a2)	-0.017 +/- 0.013
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	#2 - [359, 147]
Optimal Aperture	4.28
Optimal Annulus	5.35
Transit Duration (day)	0.068 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.108 $\pm$ 0.014 vs 0.1254 (deviation 1.24 σ)
Duration fit vs published	0.068 d vs 0.0746 d (deviation 0.22 σ)
Mid-time O–C	-0.2 min
Depth SNR	7.7
Residual scatter	0.71 %

Light curve

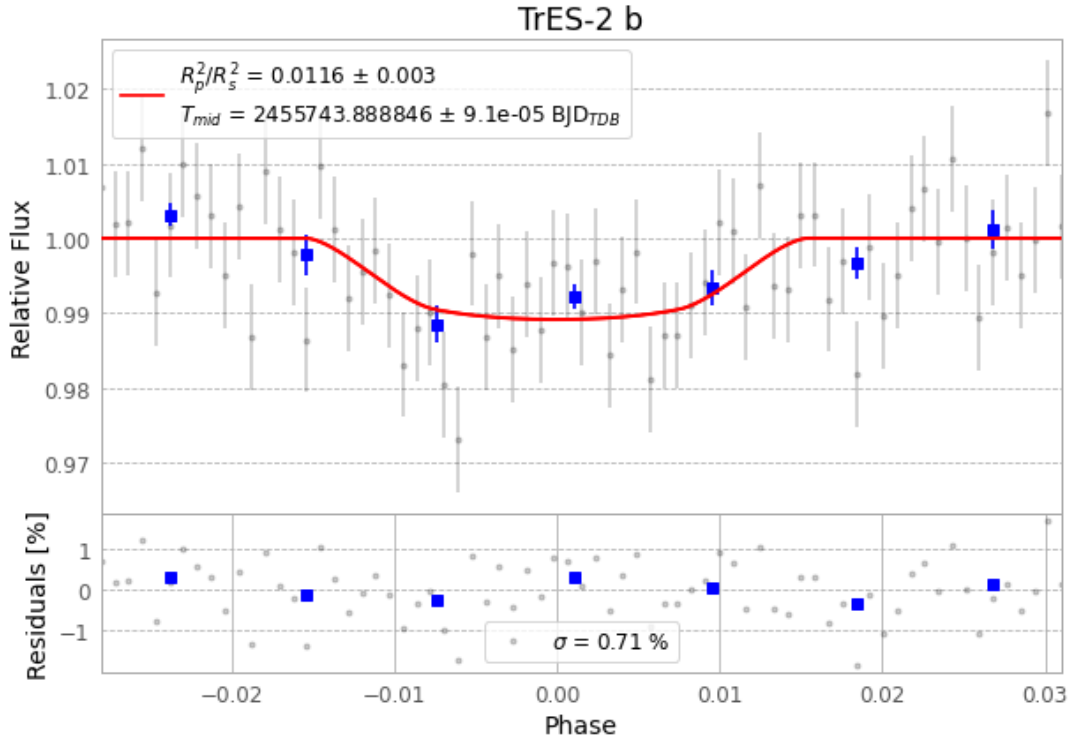


Figure 1 — FinalLightCurve_TrES-2 b_2011-07-01.png (SHA-256 a4b086cc66fc5e5e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 212], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 39ea0b93597a23bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

71 FITS frames (47 MB), TRES-2110701073613.FITS → TRES-2110701110613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110701.log (2e9cd239ff97...), FinalLightCurve_TrES-2 b_2011-07-01.png (a4b086cc66fc...), FinalLightCurve_TrES-2 b_2011-07-01.csv (b87312a3bc71...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:30Z.